

Vitamins

INTRODUCTION

Historically, vitamin requirements (or recommendations) and dietary concentrations were expressed on an international unit (IU) basis. This was done because early assays relied on biological response and did not directly measure mass of a compound and because multiple forms and sources of many vitamins are available, and they yield different potencies when evaluated using bioassays. The conversion factors for calculating IU for various forms of vitamins are in Table 8-1. These standard conversion factors have been used in previous editions.

FAT-SOLUBLE

VITAMINS

Vitamin A and p-Carotene

Sources

Vitamin A can be provided by preformed supplemental vitamin A or by the enzymatic conversion of consumed carotenoids (predominantly β -carotene). The common forms of supplemental vitamin A used in the United States are all-trans retinyl acetate and all-trans retinyl palmitate, and vitamin A activity is defined in retinol equivalents. Retinol is not found in plants, but many feeds contain β -carotene (Lindqvist et al., 2012, 2014; Pickworth et al., 2012). Carotenoids other than β -carotene can be converted to vitamin A by animals, but conversion efficiency appears to be poor, and most common feeds do not contain substantial amounts of those carotenoids. Forages can contain substantial amounts of β -carotene but most grains and grain by-products are practically void of p-carotene. Com silage contains extremely variable concentrations of β -carotene depending on duration of storage, amount of grain in the silage, and other factors (Pickworth et al., 2012). P-Carotene concentrations decrease as forages mature (Park et al., 1983). β -Carotene

decrease quickly so that stored forages (silage and hay) have lower concentrations of β -carotene than fresh forage (Bruhn and Oliver, 1978; Park et al., 1983). Under ideal wilting and ensiling conditions, loss of p-carotene for legume and grass mixtures averaged 15 percent but was as high as 25 percent (Lindqvist et al., 2012). The length of time forages are stored is negatively correlated with P-carotene concentrations (Bruhn and Oliver, 1978; Pickworth et al., 2012).

Bioavailability and Factors Affecting Supply

Bioavailability of vitamin A is defined as the proportion of vitamin A consumed that is absorbed into the body and is available to cells, but for ruminants, absolute bioavailability data do not exist. Bioavailability of vitamin A depends on the degree of ruminal destruction and on absorption efficiency by the small intestine. Based on in vitro data and data with non-ruminants, bioavailability for vitamin A is probably substantially less than 100 percent. Ruminal destruction of vitamin A can be extensive; approximately 60 percent of supplemental vitamin A was destroyed in the rumen when steers were fed hay and corn grain diets (Warner et al., 1970). Similar values have been obtained using in vitro rumen systems (Rode et al., 1990; Weiss et al., 1995). In vitro ruminal destruction of vitamin A was approximately 20 percent with high-forage diets but increased to about 70 percent with 50 to 70 percent concentrate diets. In vitro and in vivo studies suggest that between about 0 and 50 percent of dietary p-carotene is destroyed in the rumen (Potanski et al., 1974; Fernandez et al., 1976; Noziere et al., 2006), but the p-carotene contained within forages is more resistant to ruminal degradation than is supplemental P-carotene. Essentially no reliable data are available on the intestinal absorption of dietary retinyl esters in cattle, but data collected from humans and rats suggest 20 to 60 percent of it is absorbed (Blomhoff et al., 1991; Harrison, 2005). Prior to absorption, the esters are cleaved and then retinol is absorbed by what appears to be diffusion. Absorption of retinol is enhanced by increased intake of fat.

TABLE 8-1 Factors for Converting Common Sources of Vitamins into IU

Vitamin	Source	Mass/1 IU	
		Standard	Used in This Edition
A	β -carotene	2.5 microgram	2.5 microgram ^a
	All-trans retinol	0.3 microgram	0.3 microgram
	All-trans retinyl acetate	0.344 microgram	0.344 microgram
	All-trans retinyl palmitate	0.550 microgram	0.550 microgram
D	Cholecalciferol (vitamin D ₃)	0.025 microgram	0.025 microgram
	Ergocalciferol (vitamin D ₂)	0.025 microgram	0.025 microgram ^b
E	RRR-tocopherol	0.67 milligram	0.45 milligram ^b
	RRR-tocopheryl acetate	0.74 milligram	0.50 milligram ^b
	RRR-tocopheryl succinate	0.83 milligram	0.55 milligram ^b
	All-rac tocopherol	0.91 milligram	0.91 milligram
	All-rar tocopheryl acetate	1.00 milligram	1.00 milligram
	All-rat: tocopheryl succinate	1.12 milligram	1.12 milligram

^aPotency is likely overestimated using this conversion factor but data are inadequate to quantify (see text).

^bBased on cattle data, the RRR forms of vitamin E are more potent than the standard conversion factors (see text).

Apparent absorption of β -carotene from a variety of forages averaged 77 percent in dairy steers (Wing, 1969), but Cohen-Fernandez et al. (1976) reported that fecal recovery (indigestibility) of radiolabeled β -carotene was about 90 percent in sheep. In human and rodent models, intestinal absorption of (3-carotene is a saturable process (von Lintig, 2010), suggesting that absorption efficiency may be less when cows are consuming large amounts of (3-carotene (e.g., grazing) compared with cows consuming hay-based diets.

In addition to ruminal metabolism, the bioavailability of β -carotene as a source of vitamin A depends on the efficiency of converting it to retinol. β -Carotene is predominantly converted to retinol by an enzyme located in intestinal mucosal cells. Retinoic acid, at least in humans and rodents, regulates this conversion process; animals with high concentrations of retinoic acid convert less β -carotene into retinol. If this regulation occurs in cattle, cows in good vitamin A status will convert less β -carotene into retinol, which means tissue concentrations of β -carotene may increase when P-carotene is fed, whereas cows in low vitamin A status may convert much of the β -carotene they absorb into retinol and have lower β -carotene concentrations in tissues.

The vitamin A activity of β -carotene for cattle is in Table 8-1. Previously, the conversion efficiency used for humans was much greater than that used for cattle, but the assumed efficiency for humans has been reduced and is now 1 mg β -carotene (in food) = 277 IU of vitamin A or 83 μ g retinol (IOM, 2000a), which is less than the value used for cattle (1 mg β -carotene = 400 IU of vitamin A). Absorption of β -carotene in fruits and vegetables by humans was much less than previously thought (IOM, 2000b). The defined activity of P-carotene for cattle is based largely on experiments using lambs fed corn silage (Martin et al., 1968). Studies are needed to reevaluate the conversion efficiency of cattle

in light of the changes made to the human conversion efficiency factor.

Vitamin A is less stable than many other vitamins. When supplemental vitamin A is mixed in premixes without added trace minerals, loss of activity during storage under ideal conditions is similar to other vitamins at about 3.5 percent per month, but if the premix contains supplemental inorganic trace minerals (copper, iron, manganese, selenium [Se], and zinc), loss in activity was about 9 percent per month (Shurson et al., 2011). Pelleting and extrusion cause very substantial losses in vitamin A activity, and improper environmental conditions during storage (e.g., heat and exposure to sunlight) will increase loss of activity during storage (Coelho, 2002).

Relative bioavailability of vitamin A supplements can be evaluated by monitoring retinol concentrations in liver, and significant differences were found between commercial sources of supplemental vitamin A when fed to feedlot cattle (Alosilla et al., 2007). These differences could be caused by loss of activity during storage or differences in ruminal destruction or intestinal absorption. The relative dose-response assay has been used to assess dietary effects on vitamin A bioavailability and vitamin A status in calves (Hammell et al., 2000) and adult cattle (Westendorf et al., 1990), but it may lack adequate sensitivity. Because of the invasive nature or limited sensitivity of current assays, data on factors affecting bioavailability of vitamin A and p-carotene are limited. High supplementation rates of vitamin E (6,000 IU/d) tended to reduce vitamin A status in feedlot cattle (Westendorf et al., 1990), and 2,500 IU/d of supplemental vitamin E reduced plasma and tissue concentrations of β -carotene in grazing cattle (Yang et al., 2002). Feeding supplemental fat increased β -carotene concentration in plasma of dairy cows (Weiss et al., 1994).

Functions and Animal Responses

One specific function of vitamin A (retinaldehyde) is the production of rhodopsin (a vision pigment) that is necessary for low-light vision. However, because vitamin A (retinoic acid) is a major regulator of gene transcription, it is involved in a multitude of cellular and tissue functions, including spermatogenesis, female reproduction, fetal development, and maintenance of skeletal and epithelial tissue. It also is a major regulator of immune cell function and has profound effects on the immune system (Stephensen, 2001; Mora et al., 2008). Vitamin A status in cattle is positively related to various measures of immune function (Yano et al., 2009), and vitamin A supplementation enhances the function of different immune cells (Tjoelker et al., 1988a,b, 1990; Meyer et al., 2005). Cows that eventually developed retained fetal membranes had lower serum concentrations of retinol prepartum than did healthy cows (LeBlanc et al., 2004), and cows with retained fetal membranes had lower serum retinol concentrations postpartum (Akar and Gazioglu, 2006). In one study (LeBlanc et al., 2004), but not in another (Rezamand et al., 2007), cows that developed an intramammary gland infection in early lactation had lower serum retinol than healthy cows. Lower concentrations of plasma retinol were associated with more severe lameness in cows (Sadeghi-nasab et al., 2013). Stillborn calves, but not aborted fetuses, were deficient in vitamin A (Waldner and Blakley, 2014). Overall, the preponderance of data indicates that cows in suboptimal vitamin A status are at higher risk for numerous health disorders than cows in adequate vitamin A status.

Requirements

Inadequate data are available to establish a requirement for vitamin A, and because the P-carotene content of diets is highly variable and almost never known in commercial situations, an Adequate Intake (AI) was established for supplemental vitamin A rather than total vitamin A (see Chapter 1 for discussion regarding AI). Fresh forage (e.g., pasture) can have high concentrations of P-carotene; therefore, the amount of supplemental vitamin A needed when fresh forage is fed will be less than for cattle consuming conserved forages. The AI presented below assumes conserved forages are fed and are probably in excess of requirements for grazing cattle.

The vitamin A requirement established in 2001 (NRC, 2001) for growing heifers, dry cows, and lactating cows was 110 IU of supplemental vitamin A/kg body weight (BW) and was based on cerebrospinal fluid pressure, the presence of papillary edema of the eye, milk yield, immune function, mammary gland health, and reproduction (NRC, 2001). The requirement also incorporated expected ruminal destruction of a portion of the supplemental vitamin A when higher concentrate diets are fed.

The AI for supplemental vitamin A for growing heifers was kept at 110 IU/kg BW because of a lack of new data; however, the AI of vitamin A for replacement heifers remains especially poorly defined. Holstein steers fed a high-grain

diet that provided approximately 110 IU of supplemental vitamin A/kg BW had greater average daily gain than steers fed no supplemental vitamin A, but the vitamin A treatment was confounded with a vitamin E treatment (Salinas-Chavira et al., 2014). Feeding finishing diets (i.e., high concentrate) void of supplemental vitamin A has increased intramuscular fat deposition in beef steers and tended to decrease rib fat thickness compared with steers fed approximately 70 IU/kg BW (Gorocica-Buenfil et al., 2007). Whether the effects of low vitamin A intake on fat deposition occur in heifers fed lower-energy diets is not known. If inadequate vitamin A does affect fat deposition in growing dairy heifers, this could be detrimental to future milk yields if the fat is deposited in the developing mammary gland. In the beef study (Salinas-Chavira et al., 2014), the feeding period was approximately 170 days, and no adverse effects were reported in cattle not fed supplemental vitamin A. Because of reproduction demands, the lack of adverse effects when no supplemental vitamin A was fed to steers may not extend to replacement heifers.

In NRC (2001), the vitamin A requirement for adult cattle (lactating and dry) was set at 110 IU/kg BW. Milk represents a significant loss of retinol from the cow, with concentrations ranging from about 3 to 11 mg/kg of milk fat (Jensen and Nielsen, 1996; Jensen et al., 1999; Shingfield et al., 2005; Noziere et al., 2006). This is equivalent to approximately 0.1 to 0.4 mg/kg of milk with 3.7 percent fat, or about 1,000 IU of vitamin A/kg of milk. Using the average concentration, a cow producing 35 kg of milk with 3.7 percent fat would secrete about 10 mg of retinol into milk daily, which is equivalent to 30,000 IU, which is substantial relative to the AI (ca. 69,000 IU for a 625-kg cow). In addition, lactating cows are typically fed higher-concentrate diets than growing heifers and dry cows. These facts could argue for increased AI for vitamin A for lactating cows relative to the AI for heifers and dry cows and that the AI should be related to milk fat yield. The preponderance of available data on production, health, and reproduction indicates that approximately 110 IU of vitamin A/kg BW for lactating cows is adequate, but most studies used cows producing <35 kg of milk per day. Based on current data and expected loss of retinol in milk, the daily AI for vitamin A was set as follows:

If milk yield <35 kg/d (including dry cows),
AI = 110 IU/kg BW (Equation 8-1a)

If milk yield >35 kg/d, AI = 110 IU/kg BW + 1000 IU
x (milk yield - 35) (Equation 8-1 b)

Although lactating cows secrete substantial amounts of retinol into milk, several arguments exist for setting the AI for dry cows equal to that of lower-producing cows. First, the developing fetus requires vitamin A. Second, the concentration of retinol in colostrum is positively correlated with vitamin A intake during the dry period (Puvogel et al., 2008). Colostrum contains substantial amounts of retinol (see Chapter 12), and colostrum synthesis causes a sig-

nificant reduction in plasma concentrations of retinol (Goff et al., 2002). Calves are born with low vitamin A status, and increased concentrations of retinol in colostrum improve vitamin A status of the newborn calf (Puvogel et al., 2008). Third, cows fed 170,000 IU of vitamin A/d during the dry period and early lactation period produced more milk than cows fed no supplemental vitamin A (Oldham et al., 1991). Fourth, late-gestation dry cows with lower vitamin A status have increased risk of retained fetal membranes and intra-mammary gland infections in early lactation (LeBlanc et al., 2004). Because of a lack of new data, the AI for vitamin A for dry cows was retained at 110 IU/kg BW.

β -Carotene

β -Carotene has functions other than serving as a precursor for retinol. Responses to supplemental β -carotene by dairy cows have been inconsistent and could reflect differences in basal P-carotene intake or vitamin A status. The preponderance of studies has found no effect of supplemental β -carotene on milk production (de Ondarza et al., 2009). In an older review, Hurley and Doane (1989) reported that supplemental β -carotene (usually at 300 to 400 mg/d) improved some measure of reproductive efficiency in 12 of 22 studies, but when studies conducted only in North America were summarized, P-carotene had no effect on reproduction in 4 of 5 studies. Newer data show similar inconsistencies. Kawashima et al. (2009) reported cows that were anovulatory had lower concentrations of β -carotene in plasma, whereas Kaewlamun et al. (2011) reported that 1 g/d of supplemental β -carotene had no effect on ovarian activity and uterine involution. Some data have shown a relationship between low plasma concentrations of P-carotene prepartum and increased incidence of retained fetal membranes (Inaba et al., 1986; Akar and Gazioglu, 2006), whereas LeBlanc et al. (2004) found no relationship. β -Carotene, independent of its pro-vitamin A function, is an antioxidant and can enhance the killing ability of neutrophils (Chew, 1993). In some (Chew, 1987) but not all (Michal et al., 1994) studies, supplementing between 300 and 600 mg β -carotene/d reduced the incidence of intramammary gland infections and mastitis. These studies were conducted with cows at dry-off or peripartum cows. Newer data on effects of β -carotene on mammary gland health are limited. Supplementing 425 mg β -carotene/d to lactating cows did not affect somatic cell count (SCC) (de Ondarza et al., 2009). Jukola et al. (1996) reported increased mastitis in cows with low plasma P-carotene, whereas LeBlanc et al. (2004) reported no relationship. Considering the extremely inconsistent data with respect to β -carotene supplementation, an AI could not be established for β -carotene.

Maximum Tolerable Level for Vitamin A

Feeding approximately 500,000 IU of vitamin A/d (approximately 6.5 x current AI) during the dry period reduced

milk yield in the subsequent lactation possibly because of increased mammary cell apoptosis (Puvogel et al., 2005). Based on older data, cattle consuming approximately 1,300 IU of vitamin A/kg BW (approximately 12 x current AI) developed signs of osteoporosis (NRC, 1987). One-week-old calves fed 3 million IU of vitamin A/d for 10 days developed hyena disease (premature growth plate closure) (Takaki et al., 1996). In humans and other nonruminants, excess intakes of vitamin A can cause problems with bone metabolism, including osteoporosis (Penniston and Tanumihardjo, 2006); negatively affect immune function and increase incidence of certain infectious diseases (Field et al., 2002); and cause fetal abnormalities (Azais-Braesco and Pascal, 2000). With improved sensitivity of measurements, negative effects of excessive vitamin A for humans are being observed at much lower intakes of vitamin A than previously. For example, markers of osteoporosis in humans may develop when vitamin A intake is about twice the recommended daily allowance, whereas previously, 10 times requirement was considered necessary to see negative effects (Penniston and Tanumihardjo, 2006). Because of ruminal metabolism and multiple other differences, human toxicity data cannot be extrapolated to cows, but nutritionists should be aware that negative effects of excess vitamin A may occur at lower intakes of vitamin A than previously thought.

Vitamin D

Sources and Factors Affecting Supply

Vitamin D can be produced within the skin of most mammals, including cattle, as a result of the photochemical conversion of 7-dehydrocholesterol to vitamin D_r. In plants, ultraviolet irradiation causes photochemical conversion of ergosterol to vitamin D₂. Although some feeds contain vitamin D (Horst et al., 1984), there are almost no data on vitamin D concentrations of feeds published within the past 20 years (Kalac, 2012). Therefore, basal ingredients are assumed to be an unreliable source of vitamin D, and the AI is expressed on a supplemental vitamin D basis. Vitamin D in the form associated with plants, and vitamin D₃, the form associated with vertebrates, are both used for supplementation of diets. The biological activity of the two forms was generally considered equal in cattle; however, Littledike and Horst (1982) demonstrated reduced efficacy of the vitamin D_r form in cattle. Presumably, this is because reduced binding of vitamin metabolites to vitamin D-binding proteins in blood leads to more rapid clearance of vitamin D₂ metabolites from plasma. Vitamin D₃ was about twice as effective at elevating the concentration of 25-hydroxy vitamin D (i.e., calcidiol) in plasma of dairy cows as vitamin D₂ (Hympller and Jensen, 2010). In humans, the value of vitamin D, as a vitamin D supplement is questionable (Houghton and Vieth, 2006). Vitamin D₃ is the predominate source of supplemental vitamin D used for livestock, and the AI for supplemental vitamin D assumes vitamin D₃ will be used. If vitamin D₂ is used, supplementation rates

probably should be increased. Calcidiol is commercially available and can be used as a source of vitamin D. Most of the research has been with transition cows (see Chapter 12), and at this time, the relative activity (i.e., IU per unit mass) of calcidiol is not known.

Dietary vitamin D can be metabolized in the rumen by bacteria to inactive metabolites (Horst and Reinhardt, 1983), but the degree of this metabolism is unclear. Hymoller and Jensen (2010) reported that concentrations of vitamins D₂ and D₃ in rumen fluid (in vitro) were constant over time (up to 30 hours). However, concentrations were expressed per unit of dry matter (DM), which suggests vitamin D degradation occurred at a rate similar to DM digestion. Both forms of vitamin D followed similar time profiles.

In vivo synthesis of vitamin D₃ depends on the duration and intensity of exposure to solar radiation, and solar intensity depends on latitude, season, and cloud cover. Cattle housed outside have higher concentrations of 25-hydroxy vitamin D in plasma during the summer compared with winter (Hymoller et al., 2009; Edrington et al., 2012; Casas et al., 2015). To maintain adequate plasma concentrations of 25-hydroxy vitamin D, dairy cows (56° latitude) in June required about 90 minutes of sun exposure (centered on approximately 1300h) per day (Hymoller and Jensen, 2012). Based on human vitamin D synthesis rates, the required duration of sun exposure (assumed latitude of 40°) to obtain adequate vitamin D could be several times greater during spring and fall and unobtainable during the winter (Webb and Engelsen, 2006). Supplemental vitamin D is probably not needed during summer months for cattle that graze for several hours during daylight hours. However, as the date deviates from the summer solstice, sun exposure becomes an unreliable source of vitamin D for grazing cattle.

Physiology and Function

Vitamin D is a prohormone, a necessary precursor for the production of the calcium (Ca) regulating hormone 1,25-dihydroxy vitamin D. Absorbed vitamin D enters the circulation but is rapidly converted to 25-hydroxy vitamin D within the liver by vitamin D 25-hydroxylase, which is then released into the blood. Concentrations of vitamin D in blood are not a good indicator of status because of rapid removal, and blood levels usually are 1 to 2 ng vitamin D/mL plasma (Little and Horst, 1982). The production of 25-hydroxy vitamin D within the liver is dependent on vitamin D supply (dietary and in vivo synthesis). Thus, plasma 25-hydroxy vitamin D concentration is a good indicator of vitamin D status of an animal (Horst et al., 1994). However, in humans with high serum concentrations of 25-hydroxyvitamin D (i.e., good status), increased intake of vitamin D increased serum concentrations of vitamin D at twice the rate as the increase in 25-hydroxyvitamin D (Heaney et al., 2008).

The 25-hydroxyvitamin D circulates to the kidney, where it can be converted to the hormone 1,25-dihydroxy vitamin D.

This hormone acts to increase the active transport of Ca and phosphorus (P) across the intestinal epithelial cells and potentiates the action of parathyroid hormone (PTH) to increase bone Ca resorption. Both functions are vital for Ca and P homeostasis. The influence of vitamin D on Ca and P metabolism has been studied for decades, but vitamin D receptors are found throughout the body and regulate a multitude of genes involved in a host of functions in addition to Ca and P metabolism (Christakos et al., 2013). Vitamin D or, more precisely, 1,25-dihydroxyvitamin D has substantial involvement in maintaining and regulating immune function (Reinhardt and Hustmyer, 1987; Nelson et al., 2012). In bovine cell systems and in vivo, vitamin D regulates both innate (Nelson et al., 2010; Tellez-Perez et al., 2012; Alva-Murillo et al., 2014) and adaptive immunity (Nelson et al., 2011). Intramammary infusion of 25-hydroxyvitamin D reduced the severity of experimentally induced bacterial mastitis (Lippolis et al., 2011). However, intramuscular injections of vitamin D₃ to cows with clinical mastitis did not improve measures of mammary gland health (Shahmohammadi et al., 2014).

Renal production of 1,25-dihydroxyvitamin D is tightly regulated under most situations. Activity of 25-hydroxyvitamin D-1- α -hydroxylase of the kidney is stimulated by PTH, which is released in response to declining concentrations of Ca in blood (DeLuca, 1979). In the absence of PTH, when an animal is in positive Ca balance, 25-hydroxyvitamin D can be hydroxylated in the kidney to 24,25-dihydroxyvitamin D as a primary step in the inactivation and catabolism of vitamin D. The vitamin D catabolic enzymes also function to deactivate 1,25-dihydroxyvitamin D. These catabolic enzymes exist in tissues throughout the body. In these tissues, the catabolic pathway is generally stimulated by 1,25-dihydroxy vitamin D as a negative feedback to reduce high concentrations of 1,25-dihydroxyvitamin D in plasma (Reinhardt and Horst, 1989; Goff et al., 1992). Dietary supplementation of 25-hydroxyvitamin D, at least to peripartum cows, can overwhelm the feedback mechanism and significantly elevate plasma concentrations of 1,25-dihydroxy vitamin D (Wilkins et al., 2012; Weiss et al., 2015). Increased concentrations of 1,25-dihydroxyvitamin D under that situation resulted in some transient increases in plasma Ca concentrations but did not reduce clinical or subclinical hypocalcemia postpartum.

A low concentration of P in blood also can enhance renal production of 1,25-dihydroxyvitamin D, even when the concentration of Ca in plasma is normal or above normal (Gray and Napoli, 1983). Also, higher than normal concentrations of P in blood can inhibit renal production of 1,25-dihydroxy vitamin D, which can be a factor contributing to hypocalcemia in the periparturient cow (Barton et al., 1987).

Vitamin D deficiency reduces the ability to maintain Ca and P homeostasis, resulting in a decline for P and less often a decrease for Ca in plasma. This eventually causes rickets in young animals and osteomalacia in adults; both are bone diseases in which the primary lesion is failure to mineralize the organic matrix of bone. In young animals, rickets causes

enlarged and painful joints; the costochondral joints of the ribs are often readily palpated. In adult cattle, lameness and pelvic fracture are a common sequelae of vitamin D deficiency. Vitamin D deficiency in humans, as determined by low plasma concentrations of 25-hydroxyvitamin D, is a risk factor for numerous health disorders, including cancers, cardiovascular disease, diabetes, and immune dysfunction (Holick, 2007).

Requirements

The amount of dietary vitamin D required to provide adequate substrate for production of 1,25-dihydroxyvitamin D is difficult to define; therefore, the committee established an AI, rather than a requirement for vitamin D. Animals exposed to adequate sunlight may not require any dietary vitamin D, but this is highly dependent on the latitude, exposure time, and season. Sun-cured hay provided adequate vitamin D to prevent symptoms of vitamin D deficiency in young growing calves, but the hay made up most of the diet (Thomas and Moore, 1951). Other feeds are likely to provide inadequate vitamin D.

The movement away from pasture feeding systems and toward confinement and feeding of stored feeds and by-products has increased the need for dietary supplementation of vitamin D for dairy cows. The contribution of sunlight and sun-cured forage to the supply of vitamin D for the cow is not considered in this publication, and the AI for vitamin D is expressed as IU of supplemental vitamin D (assumed to be vitamin D₃). However, as discussed above, cattle that are grazing during the summer probably do not need supplemental vitamin D.

Horst et al. (1994) determined that plasma 25-hydroxyvitamin D concentrations below 5 ng/mL are indicative of vitamin D deficiency, and concentrations of 200 to 300 ng/mL would indicate vitamin D toxicosis. For humans, optimal plasma concentration of 25-hydroxyvitamin D is in the range of 30 to 50 ng/mL based on a variety of health outcomes (Bischoff-Ferrari, 2008). An optimal range of 25-hydroxyvitamin D plasma concentrations has not been established for cattle but likely is similar to the optimal range for humans. However, bovine macrophage function *in vitro* improved linearly as 25-hydroxyvitamin D concentrations increased up to 100 ng/mL (Nelson et al., 2010). Cattle with low exposure to sunlight and not supplemented with vitamin D generally have plasma concentrations of 25-hydroxyvitamin D less than 20 ng/mL (McDermott et al., 1985; Vinet et al., 1985; Hympller et al., 2009). Supplementation of 10,000 to 50,000 IU/d of vitamin D (ca. 15 to 75 IU/kg BW) usually (McDermott et al., 1985; Vinet et al., 1985; Nelson et al., 2016) but not always (Hympller et al., 2009) maintained plasma concentrations of 25-hydroxyvitamin D greater than 30 ng/mL. Cows in early lactation (<30 days in milk) had lower concentrations of 25-hydroxyvitamin D in plasma than cows in later lactation (Nelson et al., 2016), but whether this was a physiological response or reflected changes in intake is not known.

Under most circumstances, 10,000 IU/d (16 IU vitamin D/kg BW) should provide adequate vitamin D with respect to Ca metabolism for dairy cows during late gestation. Astrup and Nedkvitne (1987) reported that lactating cows producing about 20 kg of milk/d required about 10 IU vitamin D/kg BW to maintain normal concentrations of Ca and P in blood. These studies were conducted in Norway in winter and spring, when effective sunlight exposure should have been minimal. Effects on immunity and other health measures were not evaluated in those studies. However, Ward et al. (1971) reported that cows fed an alfalfa hay-concentrate diet receiving 300,000 IU vitamin D₃ once each week (43,000 IU/d) returned to estrus 16 days earlier than cows given no supplement. Ward et al. (1972) also demonstrated that cows receiving 300,000 IU vitamin D₃/wk had improved absorption of dietary Ca. Hibbs and Conrad (1983) summarized the results of several experiments and concluded that cows supplemented with 40,000 IU vitamin D₂/d (50 to 70 IU vitamin D/kg BW) produced more milk and generally ate more than cows fed the same diets with no vitamin D supplementation or supplemented with 80,000 or more IU vitamin D/d. Reduced milk production, which could be interpreted as the beginning of vitamin D toxicosis, was observed when cows were fed 80,000 IU vitamin D/d (120 to 140 IU/kg BW). In those studies, vitamin D₂ was used and

40.0 IU of vitamin D₂ may be substantially less active than 40.0 IU of vitamin D₃.

The previous vitamin D requirement (NRC, 2001) was set at 30 IU/kg BW for all classes of dairy cattle (approximately 20,000 IU/d for a typical Holstein cow). Based on a limited number of studies, for most cows, this rate of supplementation should maintain plasma concentrations of 25-hydroxyvitamin D at about 30 ng/mL, which appears adequate (Nelson et al., 2016). However, some lactating cows had plasma concentrations less than 30 ng/mL when the group was fed a diet formulated to provide 20,000 IU of supplemental vitamin D₃, but herds fed diets formulated to provide approximately 30,000 IU of supplemental vitamin D₃ per day consistently maintained plasma concentrations of 25-hydroxyvitamin D > 30 ng/mL (Nelson et al., 2016). In addition, newer studies have identified positive effects of vitamin D on immune function.

Therefore, the AI for supplemental vitamin D was set as follows:

For replacement heifers and dry cows: Vitamin D AI,
IU/d = 30 x BW, kg (Equation 8-2a)

For lactating cows: Vitamin D AI, IU/d = 40
x BW, kg (Equation 8-2b)

Although Ca metabolism can differ between some breeds (see Chapters 7 and 12), based on serum concentrations of 25-hydroxyvitamin D (Nelson et al., 2016) and number of vitamin D receptors in intestinal tissues (Liesegang et al.,

2008) , breed differences in vitamin D nutrition have not been shown. Additional experimentation is needed to determine optimal plasma concentrations of 25-hydroxy vitamin D with respect to immune function and diseases not directly related to Ca and P status. New data are needed to better titrate vitamin requirements.

Maximum Tolerable Level

Very little new information is available regarding the maximum tolerable level for vitamin D in dairy cattle. The maximum tolerable amount of vitamin D is inversely related to dietary concentrations of Ca and P. Short-term studies by McDermott et al. (1985) suggest that 50,000 IU vitamin D₃/d (80 IU/kg BW) is well tolerated while 250,000 IU vitamin D₃/d (400 IU/kg BW) is not. Hibbs and Conrad (1983) re-reported a slight decline in milk production when cows were fed 80,000 IU D₂/d (160 IU/kg BW). In nonruminants, the maximum tolerable level for vitamin D, is much greater than that for vitamin D₃. NRC (1987) suggested the maximal tolerable level of vitamin D is 2,200 IU/kg diet when fed for long periods (more than 60 days) and 25,000 IU/kg diet when fed for shorter periods of time. Vitamin D intoxication is associated with reduced DM intake (DMI), polyuria initially followed by anuria, dry feces, and reduced milk production. On necropsy, calcification of kidneys, aorta, abomasum, and bronchioles is evident (Littledike and Horst, 1982). Finishing beef steers fed 500,000 to 5,000,000 IU of vitamin D₃ the last 8 days of life had significantly greater concentrations of Ca in muscle, but no other negative effects were reported (Montgomery et al., 2004). The maximal tolerable dose of parenterally administered vitamin D is at least 100-fold lower than the maximal tolerable oral dose, and repeated injections can be especially toxic (Littledike and Horst, 1982).

Vitamin

E

Sources

Vitamin E is a generic name for a series of lipid-soluble compounds called tocopherols and tocotrienols. The most biologically active form of vitamin E is α -tocopherol; it is also the most common form of vitamin E found in most feedstuffs. α -Tocopherol has three chiral centers and can exist in eight stereoisomeric forms. Plants only make the RRR isomer of α -tocopherol, but chemical synthesis produces all eight isomers in equimolar concentrations. The concentration of RRR- α -tocopherol in plants is highly variable, but generally, it is associated with metabolically active tissues (i.e., leaves) and fat storage depots (oilseeds or seed germ). Forages and intact oilseeds (e.g., soybeans, canola, cottonseed) are the only feedstuffs with appreciable concentrations of α -tocopherol. Grains and oilseed meals

generally contain <6 mg α -tocopherol/kg of DM (McMurray et al., 1980), but dried distillers grains (ca. 12 percent oil) can contain up to 20 mg/kg (Winkler et al., 2007). Generally, the concentration of α -tocopherol in concentrate feeds is positively correlated with fat concentration. Whole soybeans contain between 5 and 30 mg α -tocopherol/kg DM (Seguin et al., 2010; Carrera and Seguin, 2016). Other oilseeds probably have similar variable concentrations of α -tocopherol. α -Tocopherol is labile, and roasting or heat processing and long exposure to oxygen destroy it (Francois et al., 2006).

Fresh forage can be an excellent source of α -tocopherol, but concentrations are extremely variable, ranging from about 20 to 150 mg α -tocopherol/kg DM (Tramontano et al., 1993; Lindqvist et al., 2012, 2014; Elgersma et al., 2013). Plant species (grasses tend to have higher concentrations than legumes), maturity (concentrations decrease as maturity increases), climate, and numerous other factors contribute to the variation. Wilting and ensiling decrease α -tocopherol concentrations by 25 to 65 percent (Müller et al., 2007; Lindqvist et al., 2012). Short wilting periods and practices that encourage rapid fermentation generally reduce losses of α -tocopherol when forages are stored as silage. Less data are available on α -tocopherol concentrations in corn silage, but values range from about 9 to 20 mg/kg DM (O'Sullivan et al., 2002; Weiss et al., 2009; Kalac, 2012). Hay usually has lower concentrations of α -tocopherol than hay crop silages with typical values <25 mg/kg of DM (Kalac, 2012), but some hays may contain twice that concentration (Weiss et al., 2009).

The form of supplemental vitamin E usually fed to dairy cows is all-rac- α -tocopheryl acetate. The esterified form of the vitamin is more stable than the alcohol form; expected losses in biological activity from premixes containing all-rac- α -tocopheryl acetate are 1 or 2 percent per month under most storage conditions, but extruded products containing all-rac- α -tocopheryl acetate may have storage losses of 6 percent per month (Coelho, 2002; Shurson et al., 2011). RRR- α -tocopheryl acetate (or the free alcohol form) is also available commercially as a vitamin E supplement.

Bioavailability

In vitro and in vivo experiments have shown that commercial forms of supplemental vitamin E are stable in the rumen over a wide range of diets (Leedle et al., 1993; Weiss et al., 1995; Chikunya et al., 2004; Hympller and Jensen, 2010). Data are not available on the efficiency of intestinal vitamin E absorption in ruminants, but in humans, less than 70 percent of ingested vitamin E is likely absorbed (Kayden and Traber, 1993). Efficiency of absorption increases as dietary fat concentration increases, and because cattle are usually fed diets with much less fat than typical human diets, vitamin E absorption by cattle may be less.

The United States Pharmacopeia (USP) has defined the factors to convert mass of the common types of supplemental

vitamin E into units related to bioavailability (see Table 8-1). Those conversion factors are based largely on research with laboratory rodents conducted decades ago, and newer data with humans and cattle have brought those conversion factors into question. The relative difference in conversion factors between the alcohol and ester forms within the main vitamin E form is likely correct (Hidiroglou et al., 1988, 1989) and simply represents dilution by the acetate moiety. The difference in bioactivity or bioavailability between the RRR and all-rac forms, however, is likely greater than the USP conversions indicate. Cattle (Weiss et al., 2009) as well as other animals (Lauridsen et al., 2002; Cortina et al., 2004; Jensen et al., 2006) have higher concentrations of α -tocopherol in blood and tissues when fed the RRR form of vitamin E compared with the all-rac form, even though on an IU basis, intake of vitamin was the same. The vitamin E requirement for humans in the United States assumes that only 2R isomers (i.e., SRS, SRR, RRS, and RRR) of vitamin E are biologically active (IOM, 2000b). Data with dairy cows and calves support that assumption (Eicher et al., 1997; Meglia et al., 2006; Weiss et al., 2009). This means that RRR forms of vitamin E are twice as biologically active than the all-rac forms (see Table 8-1), and that difference was used for this publication. However, feed labeling regulations require that the standard conversion factors be used for different forms of RRR-tocopherol. When using these forms of supplemental vitamin E, users will need to convert the labeled IU to units used in Table 8-1.

Functions and Animal Responses

The best understood function of vitamin E is as a lipid-soluble cellular antioxidant that is especially reactive with fatty acid (FA) peroxyl radicals. These compounds are produced by peroxidation of polyunsaturated FAs. Via this function and perhaps others, vitamin E is involved in the maintenance of cellular membranes, arachidonic acid metabolism, immunity, and reproductive function. Most of the research on dairy cow response to vitamin E supplementation has concentrated on reproduction and health measures, such as mastitis, retained fetal membranes, and metritis.

White muscle disease is a classic sign of a clinical deficiency of vitamin E, and it was prevented in preweaned calves when 50 IU of vitamin E/d were supplemented to a vitamin E-free diet based on skim milk (Blaxter et al., 1952). Dietary or parenteral supplementation of vitamin E to dairy cows during the peripartum period has consistently improved the function of neutrophils and sometimes macrophages (Hogan et al., 1990, 1992; Politis et al., 1995, 1996, 2001, 2004; Suwanpanya et al., 2007). In those studies, the amount of supplemental vitamin E fed per day during the prepartum period varied between 1,000 IU/d and 3,000 IU/d. In all studies, cows were fed stored forages. Feeding approximately 1,000 IU/d of supplemental vitamin E (usually all-rac- α -tocopheryl acetate) to dry cows

when adequate Se was supplemented reduced the prevalence of retained fetal membranes in most (Harrison et al., 1984; Miller et al., 1993) but not all (Wichtel et al., 1996) studies. When vitamin E was injected (usually in combination with Se), about half the time, there was no effect for prevalence of retained fetal membranes, and about half the time, there was a positive response (Miller et al., 1993). More recent studies have tended to be positive (Erskine et al., 1997; Bourne et al., 2008), especially when cows had low plasma concentrations of α -tocopherol prior to injection (LeBlanc et al., 2002). In the older studies, the typical treatment was a single injection of approximately 700 IU vitamin E and 50 mg Se, but in the more recent studies, 2,000 to 3,000 IU vitamin E were injected. A meta-analysis determined that vitamin E supplementation during the prepartum period significantly reduced the risk of cows having retained fetal membranes (Bourne et al., 2007).

The majority of studies evaluating effects of supplemental vitamin E on mastitis have been positive (Smith et al., 1985; Weiss et al., 1997; Valle et al., 2000; Politis et al., 2004; Chat-terjee et al., 2005; Rajiv and Harjit, 2005). Supplementation was usually between 1,000 and 3,000 IU/d during the dry period or peripartum period. Rates of new infection, SCCs, and the severity and duration of mastitis have been reduced with vitamin E supplementation. However, a study conducted in Canada (Batra et al., 1992) found that about 1,000 IU/d of supplemental vitamin E did not reduce the incidence of clinical mastitis. Based on the concentrations of Se in the plasma (<35 ng/mL), cows in that study were deficient in Se. In contrast to the positive studies, a large, replicated field study found that supplementing dry cows for approximately 60 days with 3,000 IU of vitamin E per day (control treatment provided 135 IU of supplemental vitamin E/d) significantly increased the risk of developing mastitis during early lactation (Bouwstra et al., 2010b). Most of the positive studies supplemented vitamin E at lower rates (1,000 IU/d) or at similar rates for shorter periods of time (14 to 45 days). Case definitions also differed between studies. Clinical data are lacking evaluating effects of supplemental vitamin E during later lactation on mastitis and other health measures.

Low plasma concentrations of α -tocopherol, especially during the peripartum period, have been related to increased risk of health problems, including mastitis, high SCCs, displaced abomasum, and retained fetal membranes (Weiss et al., 1997; LeBlanc et al., 2004; Nyman et al., 2008; Politis et al., 2012; Qu et al., 2013). However, Jukola et al. (1996) reported no relationships between plasma α -tocopherol concentrations and mammary gland and reproductive health measures, and Bouwstra et al. (2010a) reported that high α -tocopherol concentrations in plasma were a risk factor for increased mastitis.

Concentrations of α -tocopherol in plasma drop precipitously shortly before calving and remain low for a few days postpartum (Goff and Stabel, 1990; Weiss et al., 1990). This

coincides with a period of reduced immune function in dairy cows (reviewed by Sordillo, 2005). Vitamin E supplementation has improved various measures of immune function, especially in the peripartum cow (Hogan et al., 1990, 1992; Politis et al., 1996, 2001, 2004; Chandra et al., 2014). Supplementing 2,000 to 4,000 IU of vitamin E per day during the last 2 weeks of gestation reduced mammary gland infection rates, clinical mastitis, or SCCs compared with cows given

1,0 IU of supplemental vitamin E during that period (Weiss et al., 1997; Baldi et al., 2000). However, a field study on commercial farms (Persson Waller et al., 2007) found no benefit of supplementing 1,600 mg RRR- α -tocopherol per day (approximately 3,500 IU of vitamin E using the conversion factor discussed above) during the last 4 weeks of gestation on mammary gland health postpartum, but stillbirths were reduced.

Extremely high supplementation rates of vitamin E (generally 3,000 to 10,000 IU/d) have been used to reduce the development of spontaneously oxidized flavor in milk (Nicholson et al., 1991). More recently, high rates of vitamin E supplementation (3,000 to 11,000 IU/d) have been used to reduce milk fat depression associated with diets containing polyunsaturated oils, but results have been mixed. Vitamin E did not prevent or reduce milk fat depression induced by feeding diets with high inclusion rates of oil (>6 percent added oil) from rapeseed (Givens et al., 2003; Deaville et al., 2004). At more modest inclusions (<3 percent added oil), high rates of vitamin E supplementation have reduced but not eliminated milk fat depression (Focant et al., 1998; Bell et al., 2006; Pottier et al., 2006; O'Donnell-Megaró et al., 2012). In a short-term experiment, vitamin E did not reduce milk fat depression when oil supplementation started before vitamin E supplementation (Zened et al., 2012).

Requirements

Inadequate data are available to determine a requirement for vitamin E, but based mainly on cow health, an AI for vitamin E can be established. Many common feeds fed to dairy cows can contain appreciable concentrations of α -tocopherol, but the highly variable concentrations result in substantial uncertainty regarding basal concentrations. In addition, in commercial situations, few feeds are actually assayed for α -tocopherol. Therefore, the AI for vitamin E is expressed as supplemental vitamin E, not total dietary vitamin E. Because of the lack of new data, the AI for dry and lactating cows was the same as in NRC (2001). Dairy cows in the immediate (ca. 2 weeks) prepartum period benefit from increased supplementation of vitamin E (3.2 to 6.4 IU/kg BW); however, differences in supplementation rates make establishing an AI for peripartum cows difficult. The lowest supplementation rate that observed benefits (Baldi et al., 2000) was 3.0 IU/kg BW or about 2,000 IU/d during the last 2 to 3 weeks of gestation, which was used for the AI.

Dry cows: Vitamin E AI, IU/d = $1.6 \times \text{BW, kg}$
(Equation 8-3a)

Prepartum animals within 3 weeks of calving: Vitamin E
AI, IU/d = $3.0 \times \text{BW, kg}$ (Equation 8-3b)

Lactating cows and growing heifers, Vitamin E AI,
IU/d = $0.8 \times \text{BW, kg}$ (Equation 8-3c)

This is approximately equal to 1,000, 2,000, and 500 IU of supplemental vitamin E per day for dry, prefresh, and lactating cows, respectively. Fresh forage is an excellent source of vitamin E, and the need for supplemental vitamin E by grazing cattle is substantially less than those presented for cattle fed conserved forages. To account for increased supply of α -tocopherol when cows consume fresh forage, fresh forage was assumed to supply 35 mg/kg (50 IU/kg) more α -tocopherol than hay and silage. The requirement for supplemental vitamin E was reduced by 50 IU/d for every kilogram of fresh pasture DM consumed by a cow.

The difference between the AI for vitamin E for dry and lactating cows is mainly caused by expected differences in intake of vitamin E from basal feedstuffs and potentially reduced absorption of vitamin E by cows fed conventional dry cow diets (i.e., low-fat concentration). Based on typical DMI and average vitamin E concentrations in feedstuffs, the recommended amount of total vitamin E (supplemental plus vitamin provided by feedstuffs) is approximately 2.6 IU/kg BW during the late gestation and for lactating dairy cows. Of that amount, the basal diet will provide on average about

1.8 IU/kg BW for lactating cows (ranges from about 0.8 for cows fed diets based on severely weathered hay to about 2.8 IU/kg BW for cows fed diets based on pasture) and about 1 IU/kg BW (ranges from 0.5 to about 2.3 IU/kg BW) for dry cows. Colostrum synthesis during the immediate prepartum period increases the need for vitamin E. Cows may secrete 5 to 7.5 mg α -tocopherol/kg of colostrum (see Table 12-1 in Chapter 12). This is equivalent to 100 to 150 mg (or IU) of all-rac tocopheryl acetate per 10 kg of colostrum. However, plasma concentrations of α -tocopherol in mastectomized cows decrease markedly around calving (Goff et al., 2002), indicating colostrum synthesis is not the only reason peripartum cows require additional vitamin E.

Maximum Tolerable Level

Toxicity studies have not been conducted with ruminants, but data from rats suggest an upper limit of approximately 75 IU/kg BW per day (NRC, 1987). Lesser amounts of supplemental vitamin E (2,500 to 6,000 IU/d) fed to cattle had reduced vitamin A and (3-carotene concentrations in tissues (Westendorf et al., 1990; Yang et al., 2002). Dry dairy cows fed 3,000 IU of supplemental vitamin E per day during the

dry period (ca. 60 days) had a higher risk of having mastitis than cows fed 135 IU/d (Bouwstra et al., 2010b).

WATER-SOLUBLE VITAMINS

B Vitamins

A vitamins, with the possible exceptions of niacin, biotin, and vitamin B₁₂, are often not considered in diet formulation and are rarely supplemented because signs of B vitamin deficiencies are rarely observed in adult ruminants, and feeds and synthesis by ruminal microbes provide a substantial supply. However, changes in diet composition may have changed vitamin supply. In addition, marginal deficiency signs may be subtle and only cause biochemical perturbations and cellular dysfunction without observable clinical signs. The substantial increase in milk yields by today's dairy cows and the need to maximize metabolic efficiency likely have increased the demand for B vitamins. In addition, increasing vitamin concentrations in colostrum and milk may have benefits to the health of the calf and to humans consuming dairy products.

Ruminal Metabolism of B Vitamins

In ruminants, B vitamin supply cannot be calculated exclusively from B vitamin intake; significant synthesis and destruction of these vitamins by the ruminal microflora occur. Hunt et al. (1954) stated, "Members of the vitamin B-complex are synthesized in the rumen of the bovine, but our knowledge of the factors which affect these syntheses are rather limited." Table 8-2 illustrates the great variability of intake, duodenal flow, and apparent synthesis of B vitamins in rumen of dairy cows. Negative values for apparent ruminal synthesis indicate that the amount of vitamin destroyed in the rumen is greater than the amount of vitamin ingested. Absorption of B vitamins across the rumen wall has been demonstrated when the rumen is emptied of its content and filled with an aqueous solution of vitamins (Rerat et al., 1958), but in fed animals, no ruminal absorption of B vitamins is detectable (Rerat et al., 1959). As B vitamin absorption takes place mostly in the small intestine, duodenal flow of B vitamins represents the amount of vitamins potentially available for absorption by the cow. Overall, because of analytical challenges and other issues, current estimates of B vitamin synthesis, degradation, and absorption need to be improved to increase the ability to accurately determine when supplementation is warranted and will elicit a positive response.

Thiamin (B₁)

The main forms of thiamin are free thiamin and its mono-, di-, and triphosphorylated forms. Thiamin diphosphate is essential for carbohydrate metabolism (pyruvate dehydroge-

TABLE 8-2 Intake, Duodenal Flow, and Apparent Ruminal Synthesis^a of B Vitamins in Dairy Cows (mg/kg of DMI)^b

	Intake	Duodenal Flow	Apparent Synthesis in Rumen
Thiamin	1.3 to 3.8	0.8 to 7.8	-1.5 to 4.2
Riboflavin	4 to 106	3 to 87	-50 to 29
Niacin	22 to 170	47 to 146	-123 to 120
Vitamin B ₆	2.6 to 17.6	0.7 to 7.7	-14.1 to 1.3
Biotin	0.2 to 7.0	0.2 to 6.6	-0.9 to 0.2
Folates	0.2 to 1.1	0.9 to 2.4	0.5 to 1.7
Vitamin B ₁₂	^c	0.1 to 4.8	0.1 to 4.8 ^d

^aA negative value indicates that the amount of vitamin degraded in the rumen is greater than the amount of vitamin ingested.

^bSteinberg and Kaufman, 1977; Breves et al., 1981; Santschi et al., 2005a; Schwab et al., 2006; Lebzién et al., 2006; Niehoff et al., 2013; Beaudet et al., 2016; Castagnino et al., 2016a,b, 2017. No data available on pantothenic acid.

^cUnder or close to the level of detection.

^dDietary concentrations of cobalt: 0.17 to 2.5 mg/kg DM.

nase and two transketolases in the pentose-phosphate pathway), energy metabolism (α -ketoglutarate dehydrogenase in the Krebs cycle), and catabolism of branched-chain amino acids (AAs; branched-chain α -ketoacid dehydrogenases). Thiamin triphosphate is required by a peroxisomal enzyme complex for FA oxidation. Thiamin is involved in regulation of the immune system, acts as an anti-inflammatory factor, and has antioxidant properties (Manzetti et al., 2014).

Given the importance of glucose as an energy supply for the brain and because thiamin is intricately involved in several of the energy-producing reactions, thiamin deficiency causes central nervous system disorders. Polioencephalomalacia (PEM) is the most common thiamin deficiency disorder. Clinical signs include a profuse but transient diarrhea, listlessness, circling movements, opisthotonus, and muscle tremors. If treated promptly by parenteral injections of thiamin (2 mg/kg BW), the condition can be reversed (NASEM, 2016). Thiamin deficiency has been observed when thiaminases, associated with either feeds or produced from altered ruminal fermentation, destroy thiamin or produce antimetabolites of the vitamin that block thiamin-dependent reactions (Combs, 2012). Thiaminases have been detected in bracken ferns and some raw fish products. Feeding high-sulfate diets can also cause a thiamin deficiency (Gould et al., 1991), increases the need for thiamin diphosphate by the brain, and increases the risk of developing PEM (Amat et al., 2013). Thiamin is generally considered atoxic. In three short-term (3- to 4-week periods) experiments, supplemental dietary thiamin, at doses of 150 and 300 mg/d, increased milk yield in one experiment, increased milk protein yield in two experiments, and did not affect, increase, or decrease milk fat yields (Shaver and Bal, 2000). Supplementation of thiamin

in low fiber diets was more positive than when diets contained adequate fiber.

Sources of thiamin include grains, grain by-products, soybean meal, and brewer's yeast. Thiamin concentrations in rumen contents (Tafaj et al., 2004, 2006), duodenal flow (Breves et al., 1981), and apparent ruminal synthesis of the vitamin (Schwab et al., 2006; Castagnino et al., 2016a,b) are negatively correlated with ruminal pH. Between 52 and 68 percent of dietary supplemental thiamin escaped destruction in rumen (Zinn et al., 1987; Santschi et al., 2005a).

Riboflavin (B₂)

Riboflavin is the essential component of two coenzymes, flavin adenine dinucleotide (FAD) and flavin mononucleotide, involved with more than 100 enzymes in oxidation-reduction reactions. The coenzymes are essential for catabolism of certain AAs and purines, (3-oxidation of FAs, and dehydrogenation of succinate into fumarate in the Krebs cycle. Riboflavin is also involved in the reduction of oxidized glutathione (glutathione reductase) and in the activation of pyridoxine (vitamin B₆) and folates into their coenzyme forms (Combs, 2012).

Deficiency symptoms have been described in very young milk-fed calves (Wiese et al., 1947), but no deficiency or toxicity symptoms have been reported in adult ruminants. A single intramuscular injection of riboflavin (10 mg/kg for calves and 5 mg/kg for mature cows) increased neutrophil counts and enhanced neutrophil function (Osame et al., 1995). The effects of supplemental riboflavin on lactation performance have not been studied. Forages are good sources of riboflavin, although it is rapidly destroyed by sun-drying. Almost all of the riboflavin in dietary supplements (99 percent) is destroyed in the rumen (Zinn et al., 1987; Santschi et al., 2005a).

Niacin (B₃)

The generic term "niacin" covers two molecules: nicotinic acid and nicotinamide. Niacin is the essential component of nicotinamide adenine dinucleotide (NAD) and nicotinamide adenine dinucleotide phosphate (NADP), which are involved in more than 200 reactions in the metabolism of carbohydrates, FAs, and AAs and in all redox reactions. Each form has specific metabolic roles; NAD is involved in glycolysis, lipolysis, and the Krebs cycle. As such, NAD⁺ is reduced into NADH and works in synchrony with FAD, which is the ion acceptor. On the other hand, NADP is involved in the pentose-phosphate pathway and FA synthesis and acts as coenzyme of the glutathione reductase and dihydrofolate reductase. At high doses, nicotinic acid possesses antilipolytic and vasodilatory activities.

Niacin does not completely fit the definition of vitamin because in most mammals, the molecule is synthesized from tryptophan (Trp). In rats, ketone bodies (Shastri et al., 1968) and fatty liver (Fukuwatari and Shibata, 2013) suppress con-

version of Trp into niacin. The importance of endogenous synthesis of niacin differs among species (Combs, 2012). In preruminant calves, endogenous synthesis of niacin is sufficient to avoid clinical deficiency signs if the diet provides sufficient Trp (Hoppner and Johnson, 1955), but the importance of the Trp-niacin pathway for dairy cows is unknown.

Supplementation of nicotinic acid frequently increased the number of ruminal protozoa and microbial protein synthesis in vitro and in vivo (Schtissler et al., 1978; Riddell et al., 1980, 1981; Dennis et al., 1982; Shields et al., 1983; Brent and Bartley, 1984; Horner et al., 1988a,b; Erickson et al., 1990; Ottou and Doreau, 1996; Aschemann et al., 2012; Niehoff et al., 2013). According to a meta-analysis (Schwab et al., 2005) using data from 27 studies, 6 g/d of supplemental nicotinic acid did not affect lactation performance of dairy cows, but 12 g/d resulted in modest increases in yields of fat, protein, and fat-corrected milk. Feed efficiency (milk yield/DMI) tended to increase with supplemental niacin.

Supplemental niacin can have pharmacological effects on lipolysis and vasodilation, the first one to counteract the effects of lipid mobilization in early lactation and the second one to reduce the consequences of heat stress on lactating dairy cows. However, results have not been consistent. Decreases in plasma concentrations of FAs and P-hydroxybutyrate and increases in plasma glucose are the most frequently reported responses following use of nicotinic acid supplements (dose ranging from 6 to 12 g/d), although the response is highly variable among studies (Schwab et al., 2005; Niehoff et al., 2009; Pescara et al., 2010). Supplementary nicotinic acid (doses range from 12 to 36 g/d) increases vasodilation, enhancing heat loss during periods of heat stress in some studies (Di Costanzo et al., 1997; Niehoff et al., 2009; Pescara et al., 2010; Zimbelman et al., 2010, 2013; Wrinkle et al., 2012; Pineda et al., 2016) but not in others (doses varying from 4 to 24 g/d; Loholter et al., 2013; Rungruang et al., 2014). In nonruminant animals, toxicity of niacin is low, at least 10- to 20-fold the estimated requirements (Combs, 2012).

Brewer's yeast and distillers grains are good sources and forages are considered fair sources of niacin (McDowell, 2000). Concentrations of niacin in cereals are often high, but a large proportion is covalently linked to small peptides and carbohydrates, which markedly impairs its availability, at least in nonruminant animals (Combs, 2012). Availability of those complexes to ruminants is not known. Destruction in rumen of supplementary niacin, given as nicotinic acid or nicotinamide, is greater than 90 percent (Zinn et al., 1987; Santschi et al., 2005a). However, production responses to supplementation of rumen-protected (RP) forms of niacin have been small (Yuan et al., 2012; Pineda et al., 2016).

Pantothenic Acid (B₅)

Pantothenic acid is an essential component of coenzyme A (CoA) and the acyl carrier protein (ACP). ACP is at the center of the multienzyme complex, FA synthase, and as such, its

component, 4'-phosphopantetheine, acts as an arm to allow the binding and transfer of acyl units for the elongation of the FA chain. Coenzyme A (CoA) is essential for numerous enzymatic reactions within cells, including the Krebs cycle, lipid metabolism, and AA catabolism, and acts as a global regulator of energy metabolism. CoA cannot pass through cell membranes, but all tissues can synthesize it using pan-tothenic acid. Conservation of CoA within cells is due to a tight control on CoA synthesis but also to efficient recycling of phosphopantetheine formed during catabolism of CoA and ACP (Bender, 1999).

Deficiency symptoms have been described in calves fed a pantothenic-free synthetic milk (Sheppard and Johnson, 1957), but no deficiency or toxicity symptoms have been reported in adult ruminants. Dietary supplementation of 1 g/d pan-tothenic acid decreased the efficiency of ruminal microbial protein synthesis of cows fed a low-forage diet, whereas it increased the amount of organic matter ruminally fermented with a high-forage diet (Ragaller et al., 2011). In the same study, pantothenic acid decreased plasma glucose with the low-forage diet and decreased milk protein content and increased lactose content with the high-forage diet. In a field study, supplements of pantothenic acid protected (50, 100, or 200 mg/d) or not protected (200 mg/d) from degradation in rumen increased milk production, milk fat and protein contents, and plasma concentration of glucose in cows during the first 5 months of lactation (Bonomi, 2000). However, supplementation of unprotected pantothenic acid (21 mg/kg DM) fed alone or in combination with biotin (0.87 mg/kg DM) for 18 days had no effect on DMI and yields of milk and milk components (Ferreira et al., 2015).

Pantothenic acid, usually in its bound forms (CoA, CoA esters, ACP), is widely present in feed ingredients from plant and animal origins. In sheep, the amount of free pantothenic acid reaching the duodenum is positively correlated with its intake, whereas the amount of CoA reaching the duodenum is positively correlated with the amount of microbial DM synthesized in the rumen (Finlayson and Seeley, 1983). In steers, only 22 percent of supplemental pantothenic acid escaped degradation in the rumen (Zinn et al., 1987). Similarly, only 15 percent of a pantothenic acid supplement was not degraded in an artificial rumen (Volker et al., 2011).

Vitamin B₆

There are six vitamers with vitamin B₆ activity: pyridoxine, pyridoxamine, and pyridoxal and their respective phosphorylated forms. Pyridoxal-5-phosphate (P-5-P) is a coenzyme for more than 120 enzymes and is involved in most reactions in AA metabolism. Due to its critical roles in AA metabolism, vitamin B₆ requirements of nonruminants are increased by high-protein diets (Okada et al., 1998). The vitamin is also essential for glycogen utilization; synthesis of histamine, hemoglobin, and sphingolipid; and modulation of expression of some genes. In nonruminants, symptoms

of deficiency are nonspecific neurologic and dermatologic changes. There is no report of deficiency symptoms in adult ruminants. The effects of vitamin B₆ supplementation on lactation performance of dairy cows have not been studied. In nonruminants, toxicity of vitamin B₆ is low, although it is neurotoxic at excessively high doses, over 1,000 times the reference nutrient intake (Bender, 1999; Combs, 2012).

Forages and grains are good sources of vitamin B₆, but diet composition does not have a major effect on vitamin B₆ ruminal concentrations (Kon and Porter, 1953; Briggs et al., 1964; Lardinois et al., 1994; Santschi et al., 2005b) probably because apparent ruminal synthesis of vitamin B₆ is negatively correlated with B₆ intake (Beaudet et al., 2016; Castagnino et al., 2016a,b, 2017). However, 60 to 100 percent of supplemental B₆ escaped destruction in rumen (Zinn et al., 1987; Santschi et al., 2005a).

Biotin

Biotin plays key roles in lipid, AA, and energy metabolism due to its function as coenzyme for five carboxylases that catalyze the incorporation of the most oxidized form of one-carbon units (i.e., bicarbonate). Two of these carboxylases (pyruvate carboxylase and propionyl-CoA carboxylase) are likely of major importance for ruminants due to their role in gluconeogenesis. Methylcrotonyl-CoA carboxylase is involved with the catabolism of leucine (Leu), and two forms of acetyl-CoA carboxylase (mitochondrial and cytosolic) are involved with FA synthesis and oxidation. Biotin is involved in regulation of gene expression of many enzymes that play critical roles in glucose metabolism.

In many species, the major sign of a biotin deficiency is skin lesions. In vitro, omission of biotin from the culture media markedly reduces ruminal cellulose digestion and volatile FA production, especially propionate (Milligan et al., 1967). However, biotin supplementation has not improved in vitro and in vivo fiber digestibility (Majee et al., 2003; Rosendo et al., 2003). Two meta-analyses (Chen et al., 2011; Lean and Rabiee, 2011) evaluated the effects of supplemental biotin on milk production (some data were used by both analyses) with similar conclusions. Biotin supplements, at a dose of 20 mg/d, increased DMI, milk production, and fat and protein yields but did not affect milk fat and protein concentrations. Numerous studies report an improvement in hoof health when 10 to 20 mg/d supplemental biotin is fed (Lean and Rabiee, 2011). High doses of biotin are considered atoxic.

Yeast is a good source of biotin, and oilseed meals contain more biotin than cereals, with corn being a better source than wheat and barley. In feeds, biotin is present as free biotin and as biocytin, biotin bound to protein lysyl residues by an amide link. This bond can only be broken by the enzyme, biotinidase, present in intestinal mucosa and pancreatic juice. Biotinidase is rarely used for sample preparation because no pure preparation of the enzyme is available commercially; therefore, differences in extraction methods leading to incomplete liberation of free

biotin exacerbate the variability among studies. Duodenal flow of biotin is related to the amount of fermented organic matter and microbial protein synthesis (Lebzien et al., 2006). Bioavailability of dietary supplements of biotin has been estimated around 45 percent (Frigg et al., 1993; Santschi et al., 2005a).

Folates

Folic acid is used either as the generic name of the vitamin or, specifically, for the synthetic form of the vitamin, pteroylmonoglutamic acid. The term “folates” applies to the numerous biologically active forms: dihydrofolate and several forms of tetrahydrofolate. The length of the glutamate chain can vary from one to seven glutamate molecules. In mammals, folic acid accepts and releases one-carbon units in biochemical reactions. Cellular tetrahydrofolate accepts one-carbon units from donors such as serine or formate and transfers them for thymidylate and purine synthesis. Therefore, folic acid is crucial for DNA synthesis, replication, and repair. A folic acid deficiency causes an imbalance in DNA precursors, uracil misincorporation, and chromosome breakage. Tetrahydrofolate can also transfer methyl groups to homocysteine for regeneration of methionine (Met) under the action of a vitamin B₁₂-dependent enzyme, Met synthase. In the methylation cycle, the role of folate coenzymes is to provide one-carbon units to ensure a constant supply of S-adenosylmethionine, which is the primary methylating agent. Reactions mediated by S-adenosylmethionine include DNA methylation, which controls gene transcription and genetic stability, as well as synthesis of phosphatidylcholine, choline, creatine, and several neurotransmitters.

Weekly intramuscular injections of 40 mg folic acid given to dairy heifers from 10 days until 16 weeks of age increased average daily gain by 8 percent during the 5 weeks following weaning (Dumoulin et al., 1991), suggesting that folic acid may be deficient in young calves around weaning when the ruminal microbial populations are not fully established. Daily dietary supplements of folic acid (2 to 6 mg/kg BW of unprotected folic acid or 1 to 3 g of a RP product) usually (Girard and Matte, 1998; Graulet et al., 2007; Girard et al., 2009a; Li et al., 2016) but not always (Girard et al., 2005) increase milk production and milk protein yield during the first part of the lactation. Except for one study (Li et al., 2016), none of these studies observed an increase in DMI, suggesting that supplemental folic acid increases metabolic efficiency. Li et al. (2016) also reported improved reproductive efficiency when cows were supplemented with RP folic acid. Dietary supplements of folic acid have little effects on ruminal fermentation (Chiquette et al., 1993; Girard et al., 2009a; Ragaller et al., 2010). High doses of folic acid have no negative effects in nonruminant animals, except in the presence of vitamin B₁₂ deficiency (Selhub et al., 2007; Combs, 2012).

Oilseeds and brewer's yeast are major dietary sources. Disappearance of supplementary folic acid before the duo-

denal cannula is 97 percent but 25 percent of a dose of folic acid infused in the abomasum disappears before the duodenal cannula, probably absorbed in the proximal part of the duodenum (Santschi et al., 2005a). Based on the latter, destruction of a dietary supplement of folic acid can be estimated around 72 percent of the amount ingested.

Vitamin B₁₂

“Vitamin B₁₂” is a generic term used to describe all corrinoids containing an atom of cobalt (Co) and exhibiting the biological activity of cyanocobalamin. Cyanocobalamin is the synthetic form of vitamin B₁₂ present in most supplements. The cyanide group is added to stabilize the molecule, but the molecule is not biologically active until the cyanide group is enzymatically removed. In mammals, the major cobalamin vitamers are methylcobalamin, adenosylcobalamin, and hydroxocobalamin.

Several vitamin B₁₂-dependent metabolic reactions have been identified in microorganisms, but in mammals, only two such reactions exist. One of the two vitamin B₁₂-dependent enzymes, Met synthase, is the critical interface between folic acid and vitamin B₁₂ metabolism. Met synthase transfers a methyl group from 5-methyl-tetrahydrofolate (producing tetrahydrofolate) to homocysteine producing Met. In a vitamin B₁₂ deficiency, all available one-carbon units are diverted toward the synthesis of 5-methyl-tetrahydrofolate, which cannot be demethylated by Met synthase in absence of vitamin B₁₂, leading to a secondary folate deficiency. Besides its role in the methylation cycle and folate metabolism, vitamin B₁₂ plays a key role for the entry of propionate in the Krebs cycle and gluconeogenesis, through the mitochondrial vitamin B₁₂-dependent enzyme, methylmalonyl-CoA mutase.

Vitamin B₁₂ deficiency has been demonstrated in pre-ruminant calves fed diets devoid of animal protein (Lassiter et al., 1953). In ruminants, vitamin B₁₂ deficiency is the major consequence of an insufficient supply in Co. However, even with sufficient dietary Co, plasma concentrations of vitamin B₁₂ are low during the first weeks of lactation (El-liot et al., 1965; Mykkanen and Korpela, 1981; Girard and Matte, 1999; Kincaid and Socha, 2007). Dietary or parenteral supplemental vitamin B₁₂ when cows are fed adequate Co has minor or no effects on production responses (Elliot et al., 1979; Croom et al., 1981; Kincaid and Socha, 2007; Grace and Knowles, 2012; Akins et al., 2013). However, a combined supplement of folic acid and vitamin B₁₂ given from 3 weeks before calving until 8 or 16 weeks of lactation (usually given parenterally once weekly) increased milk production and energetic efficiency in early lactation (Girard and Matte, 2005; Graulet et al., 2007; Preynat et al., 2009a,b; Ghaemialhashemi, 2013; Duplessis et al., 2014a; Gagnon et al., 2015). Perhaps via improved energy status, the combined vitamin supplement has improved various measures of reproductive efficiency (Ghaemialhashemi, 2013;

Duplessis et al., 2014b; Gagnon et al., 2015). No toxicity of vitamin B₁₂ has been reported. Parenteral supplementation of a commercial mixture of butophosphan (an organic P compound) and vitamin B₁₂ before calving or in early lactation decreased plasma concentrations of nonesterified FAs and β -hydroxybutyrate (Furll et al., 2010; Rollin et al., 2010; Kreipe et al., 2011; Pereira et al., 2013; Nuber et al., 2016). Production responses to that mixture are variable.

Vitamin B₁₂ is not synthesized by plants; it is produced only by bacteria and archaeobacteria when Co supply is adequate (Martens et al., 2002). Only 11 percent of the Co ingested is used for ruminal synthesis of corrinoids, of which only 4 percent is incorporated into vitamin B₁₂ (Girard et al., 2009b). Ruminal bacteria use dietary Co to produce vitamin B₁₂ analogues, which are devoid of biological activity. The production of biologically active vitamin B₁₂ usually increases with Co intake, generally at the expense of analogue synthesis (Hedrich et al., 1973; Bigger et al., 1976; Tiffany et al., 2003, 2006; Stemme et al., 2008). Apparent ruminal synthesis of vitamin B_p is correlated positively with Co intake (Beaudet et al., 2016; Castagnino et al., 2016b, 2017). Apparent ruminal synthesis of vitamin B₁₂ in the rumen is generally positively correlated with fiber intakes (Sutton and Elliot, 1972; Schwab et al., 2006; Beaudet et al., 2016; Castagnino et al., 2016a,b, 2017) and negatively correlated with the amount of starch digested in rumen (Sutton and Elliot, 1972; Schwab et al., 2006; Beaudet et al., 2016).

Choline

Choline is not a vitamin in a traditional sense because it can be synthesized by cows (i.e., not dietary essential) and it is required in gram rather than milligram or microgram amounts. Johnson et al. (1951) produced a choline deficiency in week-old dairy calves using synthetic milk replacer diets containing 15 percent casein. Choline requirements estimated from that experiment were 260 mg/L of milk replacer (1,733 mg/kg DM). Current estimates of requirements for the calf are 1,000 mg/kg DM. The predominant sign of choline deficiency in most animals is fatty liver; in calves, other deficiency signs include muscular weakness and renal hemorrhage.

Both naturally occurring choline in feeds, predominantly found in phospholipids (lecithin), and dietary choline from supplements such as choline chloride are extensively degraded in the rumen (Neill et al., 1979; Sharma and Erdman, 1988a,b). Microbial degradation of choline in the rumen results in the production of acetaldehyde and trimethylamine. Methyl group carbon from trimethylamine is subsequently degraded to methane (Neill et al., 1978). Supplementation of choline in an unprotected form is useless because of extensive ruminal degradation. Because of extensive degradation of dietary choline, methyl groups for synthesis of methyl-containing metabolites in the dairy cow are presumably produced via

methylation pathways involving Met and the enzyme, S-adenosylmethionine methyl transferase. Sources of methyl groups for ruminants would include intestinally absorbed Met, betaine resulting from degradation of choline, and de novo synthesized methyl groups produced through 5-methyl tetrahydrofolate. Approximately one-third of the Met methyl groups were transferred to choline in studies with lactating dairy goats (Emmanuel and Kennedy, 1984). Intravenous infusion of choline and carnitine reduced the irreversible loss of Met by 18 to 25 percent in sheep, suggesting that Met could be spared with the addition of methyl-group-containing metabolites (Lobley et al., 1996).

Choline concentration of milk ranges from about 70 to 100 mg/L (Deuchler et al., 1998; Pinotti et al., 2003; Elek et al., 2008) and increases 25 to 40 percent when RP choline is fed (ca. 15 g/d actual choline). This suggests that secretion of choline into milk could be qualitative indicator of post-ruminal choline supply. Daily excretion rates vary from about 2 to 6 g/d depending on milk yield and whether RP-choline was fed.

The dairy cow evolved under circumstances where intestinally absorbed choline is almost nonexistent; hence, choline supply is dependent on the ability of the cow to synthesize it from serine or from feeding RP forms. Since publication of the first article on the use of RP-choline (Erdman and Sharma, 1991), adequate studies have now been published to allow for meta-analyses. Sales et al. (2010) used data from 11 different publications and Arshad et al. (2020) used data from 20 publications, including most of the publications used by Sales et al. (2010). Results were in general agreement: supplementing approximately 13 g of actual choline (in RP form) increased milk yield about

1.5 kg/d, increased DMI about 0.5 kg/d, and had little effect on milk composition but increased milk fat and protein yields. Most studies conducted with RP-choline only involve transition cows (supplementation usually started a few weeks prepartum and usually ended 3 to 4 weeks postpartum). Data on production responses to RP-choline later in lactation are limited and inconsistent. The predicted milk yield response (in early lactation) to RP-choline was greater when calculated (not measured) supply of metabolizable Met was low and response decreased as metabolizable Met supply increased (Arshad et al., 2020). This meta-analysis and individual studies indicate that RP-choline may reduce the amount of Met being used as a methyl donor, allowing more to be used for protein synthesis. In an experiment where methyl transfer from Met was inhibited but choline was provided, fat-correct milk yield increased, suggesting the importance of Met in methyl group metabolism in the dairy cow (Sharma and Erdman, 1988b). However, in studies in which both RP-choline and RP-Met have been fed, few interactions have been observed (Sun et al., 2016; Zhou et al., 2016).

Because the role choline has in hepatic lipid metabolism (Piepenbrink and Overton, 2003), RP-choline has

been investigated as a means of reducing incidence and severity of ketosis. Although a meta-analysis (Arshad et al., 2020) found that peripartum cows fed RP-choline had significantly higher blood glucose concentrations and lower blood concentrations of FAs and P-hydroxybutyrate, the mean differences were clinically insignificant, and several individual studies report no differences. However, several studies have reported improved health of peripartum cows when RP-choline was supplemented. Improvements included fewer cases of retained placenta, less mastitis, and displaced abomasum (Ardalan et al., 2010; Lima et al., 2012). Although health data are limited, a meta-analysis (Arshad et al., 2020) reported that RP-choline significantly reduced the incidence of retained placenta and mastitis but did not affect prevalence of other health disorders, including ketosis and fatty liver. Limited data also suggest that positive health effects are more likely when cows rather than prepartum heifers are supplemented with RP-choline (Lima et al., 2012).

Supplementing peripartum cows with an effective source of RP-choline (i.e., not degraded in rumen but available for intestinal absorption) is expected in increased production measures and can reduce the prevalence of some health disorders, but the committee did not establish a dietary requirement for choline because it is synthesized by cows and because of potential variability in commercial products.

Vitamin C

Vitamin C or ascorbic acid is synthesized from L-gulonic acid within the liver of ruminants. Calves cannot synthesize ascorbic acid until approximately 3 weeks of age (Cummins and Brunner, 1991). Hence, vitamin C is not considered an essential nutrient for healthy cattle that are older than about 3 weeks. Most orally ingested ascorbic acid is destroyed in the rumen, but some commercial formulations of vitamin C may provide varying degrees of protection from ruminal metabolism. Oral supplementation of modified forms of vitamin C designed to reduce ruminal degradation has increased or tended to increase plasma ascorbic acid in sheep (Hidiroglou et al., 1997), steers (Pogge and Hansen, 2013), and dairy cows (Weiss, 2001). Ascorbic acid functions as a water-soluble cellular antioxidant and is involved in numerous biochemical pathways (Smirnov, 2018). Vitamin C is needed for collagen synthesis, iron absorption, and phagocytic cell function, among other functions. Stressful (e.g., poor housing conditions, heat stress) and inflammatory events (e.g., mastitis) reduce plasma concentrations of ascorbic acid in calves and cows (Hidiroglou et al., 1977; Cummins and Brunner, 1989; Weiss et al., 2004; Padilla et al., 2006). Clinical ketosis did not affect plasma ascorbic acid concentrations (Padilla et al., 2005). Although dietary supplementation of some forms of vitamin C can increase concentrations of ascorbic acid in blood, and lower plasma concentrations of ascorbic acid are associated with mastitis, dietary supplementation

of vitamin C to cattle has had little to no effect on diseases and immune cell function (Santos et al., 2001; Chaiyotwit-tayakun et al., 2002; Naresh et al., 2002; Weiss and Hogan, 2007). No growth response has been reported when calves were supplemented with vitamin C. Immunoglobulin titers in calves were generally not affected by vitamin C supplementation (Cummins and Brunner, 1989; Hidiroglou et al., 1995). Current data do not support routine supplementation of vitamin C to calves or adult cattle.

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